

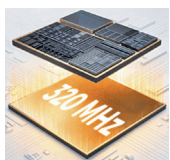
NEW PRODUCTS

TECHNOLOGY

COMPONENTS

All-in-one system-on-chip

Espressif Systems has unveiled the ESP32-S31, an all-in-one system-on-chip integrating Wi-Fi 6, Bluetooth 5.4, IEEE 802.15.4, Ethernet, edge AI, and hardware security to simplify advanced IoT device design. The dual-core



32-bit RISC-V chip runs at 320MHz, features a 128-bit SIMD path, 512kB SRAM, and supports high-speed external memory expansion. The chip enables high-performance processing and connectivity in a compact design.

Espressif Systems

<https://www.espressif.com>

Module for IoT networks

Developed by Nordic Semiconductor, the nRF93M1 Cat 1 bis module



integrates cellular connectivity with cloud-ready features, simplifying device onboarding, firmware updates, and monitoring.

Engineered for scalable IoT networks, it supports ultra-low-power operation, multiple SIM options, and standardised interfaces to ease integration with existing systems.

Nordic Semiconductor

<https://www.nordicsemi.com>

Optical chips

A new class of optical chips, Cardinal, is a family of 3nm optical DSPs built to handle the growing bandwidth,



latency, and reliability demands of AI compute networks. Designed for AI networks,

it leverages light-based data processing to deliver ultra-high speed, low latency, and energy-efficient performance for next-generation computing. Built using photonic technologies, the chips transmit and process information using light instead of electricity, significantly improving bandwidth and reducing power consumption.

Credo Technology Group Holding Ltd

<https://investors.credosemi.com>

GaN power system

A new gallium nitride (GaN) power system delivers smaller, more efficient chargers through high-frequency



operation, reducing magnetic components and heat while boosting energy efficiency and thermal performance. Developed by GaN engineers, it enables higher power density, faster multi-device charging, and reflects industry adoption, replacing silicon in next-generation electronics.

Renesas Electronics Corporation

<https://www.renesas.com/en>

Automotive MOSFET

An automotive MOSFET family 100V PowerDI 8080-5 targeting 48V vehicle architecture has been introduced to



improve efficiency, thermal performance, and power density in next-generation cars. Designed with ultra-low on-resistance and compact packaging, the devices reduce conduction losses while enabling higher current handling

and reliable operation under demanding conditions. The MOSFET family supports applications such as DC-DC converters, motor drives, and battery systems.

Diodes Incorporated

<https://www.diodes.com>

Noise filters

Miniaturised noise filters MAF-0603GWY series designed for compact electronic devices is enabling cleaner



audio output by suppressing electromagnetic interference and signal distortion in constrained form factors. These

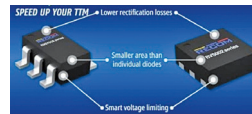
filters use combinations of inductors, capacitors, and resistive elements to block unwanted high-frequency noise while preserving desired audio signals. Developed for integration into smartphones, wearables, and portable audio systems, they address interference from RF circuits and dense PCB layouts.

TDK Corporation

<https://www.emea.lambda.tdk.com/uk>

Discrete rectifiers

A discrete rectifier solution RVS002 integrating a full-bridge design has been introduced to reduce board space and improve efficiency in compact



power systems. The device combines

MOSFETs and Schottky diodes within a single package, eliminating multiple external components used in conventional rectification stages. Developed for modern power electronics applications, the innovation reflects increasing demand for compact, efficient designs as system miniaturisation accelerates.

RECOM Power

<https://recom-power.com/en>